TRAIN SIM CLASSIC · VEHICLE WORKFLOW GUIDE

Part 3

Animations and IA export

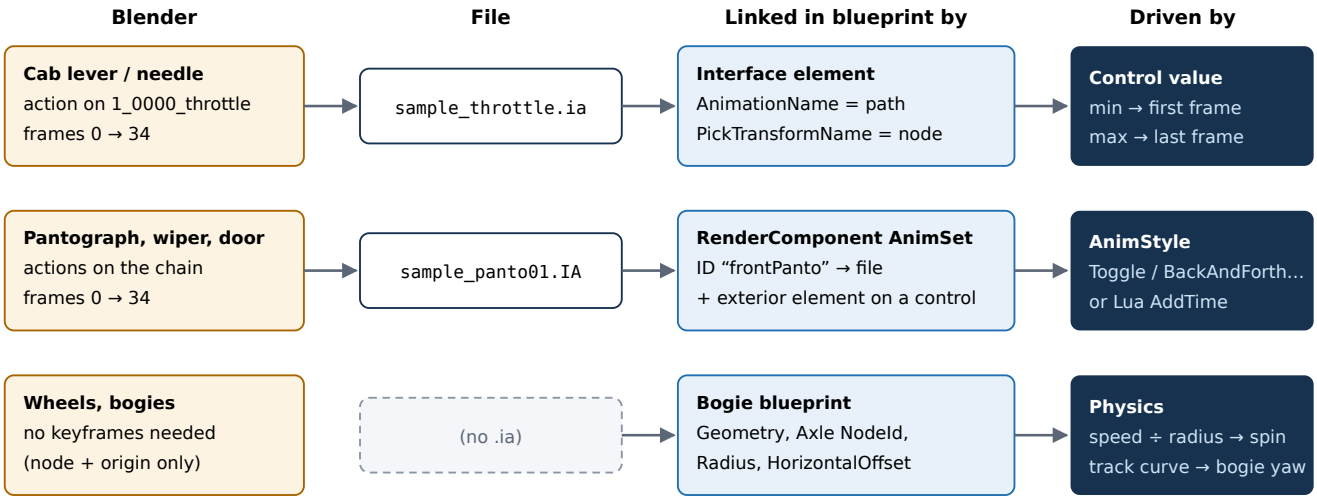
Pantographs, wipers, doors, cab levers and needles: how to key them in Blender so that one `.ia` file per animation drops straight into the blueprints and behaves the same in game as in the viewport.

In this part

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BRIAGE references are to the v2.5.786 manual. Sample data: the electric loco's Blender scene reads (12 cab actions and 6 pantograph actions, 35 frames each at 24 fps) and its blueprint.

1 · The three ways TSC moves a node



Every row ends at a name the blueprint stores. The .ia only says how nodes move; the blueprint says when.

Figure 3.1 · Three animation paths. Row A is used by every cab control in the sample (11 object actions and one armature action). Row B is used by the two pantographs and the wipers (`frontPanto` / `rearPanto` with AnimStyle `Toggle` , `Wipers` with `BackAndForthContinuous`). Row C needs no keyframes at all. **verified**

2 · What an .ia file contains

- For each exported node: **position and rotation** per frame. **No scale** (BRIAGE §8.5.2), no materials, no geometry.
- A **sample rate** = Blender fps × Frame rate Multiplier, visible as `SampleRate` in the log (BRIAGE §5.2).
- Node names that must exist in the IGS the animation is played on. The .ia and .igs are matched **by node name**, so an .ia exported from a scene whose names differ from the IGS moves nothing.
- Motion relative to each node's parent when **Relative export** is on (default, BRIAGE §5.5), which is what you want for chains like pantograph arms and for buttons mounted on moving parts.

3 · The golden rules

Seven rules that prevent almost every animation bug

1. **Frame 0 is the rest pose** and must match the exported IGS. The IGS is exported at frame 0 (BRIAGE §8.4.1).
2. **One independent motion = one .ia file.** Everything in one file moves together. Different controls need different files.
3. **Each animated node must be a separate node** in the IGS: a keyword name or a custom keyword, otherwise it is merged away (Part 2).
4. **Cab controls span min → max.** The first frame is the control's minimum and the last frame its maximum; the game scrubs linearly between them.
5. **Use LINEAR interpolation** for anything scrubbed by value (levers, needles), or the lever lags behind the value at the ends. BRIAGE also recommends linear to cure stutter (§9.6).
6. **Apply scale and rotation before keying** (BRIAGE §9.9). Never animate scale.
7. **What the viewport plays is what is exported.** Constraints, drivers, bones and child-of links are baked as evaluated (BRIAGE §8.5.1), so a disabled constraint is also disabled in the export.

Several animations in one .blend

BRIAGE offers two ways (BRIAGE §8.5.3), and the sample uses the first:

A · Same range, separate selection

All actions run 0–34. For each export, select only the objects of that animation. The sample's 12 cab actions are all 0–34. The toolkit's “Prepare next IA job” does the selecting for you.

B · Separate frame ranges

Wipers 0–100, doors 120–200, and so on. Set the scene range to the animation before each export. Enable **Trim Animation** (§5.4) to drop idle frames at either end.

4 · Worked example: pantograph

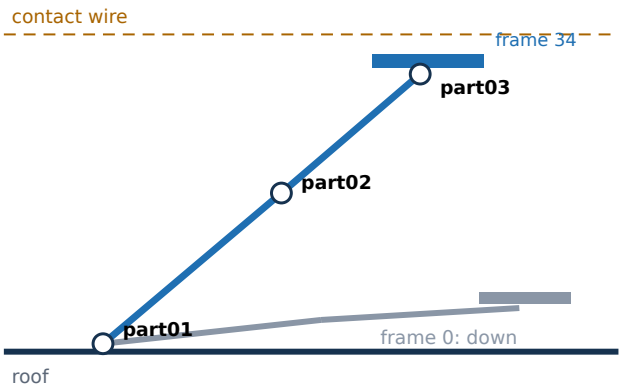
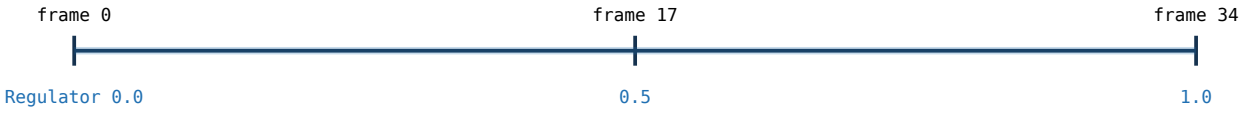


Figure 3.2 · The sample's pantograph chain. part02 is parented to part01 and part03 to part02 (verified in the geometry's transform dependencies), so each part only rotates about its own hinge.

1. Model the arms as separate objects, origins on the hinges, named `1_0128_panto01_part01..03` (`panto$$` is a keyword, so no custom keyword is needed).
2. Parent part02 → part01 and part03 → part02.
3. Key frame 0 down, frame 34 up (the sample: 35 frames at 24 fps ≈ 1.46 s). Use a rig or constraints if you like; the evaluated motion is baked.
4. Select **part01** (children follow when **Selected hierarchy** is on, §5.6) and export `sample_panto01.ia`. Repeat for panto02.
5. In BPE2: RenderComponent ▸ AnimSet ▸ add ID `frontPanto` → file; tick **Animate in editor**.
6. In ControlValues: `FrontPantographControl` gets an **Exterior animation** element, AnimationID `frontPanto`, AnimStyle `Toggle`.

5 · Worked example: cab lever and needle



Rotation keyed at frame 0 (lever back) and frame 34 (lever full forward), interpolation LINEAR.

A notched lever (e.g. TrainBrakeControl notches Release 0 … Emergency 0.95) lands at value × last frame.

A needle (SpeedometerMPH, range 0–500 in the sample) must sweep the dial so that the maximum value reads correctly at the last frame

Figure 3.3 · Value-to-frame mapping for a scrubbed cab animation. The frame count only sets the resolution; the fps is irrelevant because the game never “plays” this animation, it positions it.

- **Calibrate the dial against the control range.** The sample's `SpeedometerMPH` range is 0–500, so a dial drawn for 0–150 mph would read wrong: either change the range to the dial's full scale or key the needle to the 500 mark. verified
- **Needles beyond 180°** stutter with too few samples: use Frame rate Multiplier 7 and an odd frame count (≈25) (BRIAGE §8.5.4).
- **Bone-driven controls** (the sample's `sample_start.ia` armature with bones `2_0000_start` / `3_0000_start`) export as pose animation; make sure the matching nodes exist in

the cab IGS. In the sample they don't, which is why the start button is dead.

6 · Worked example: wipers and doors

Wipers

Key a full cycle: park → sweep → park. Tick **Remove last frame** (§5.3) so the loop does not double the park frame. Link as AnimSet `Wipers`, exterior element AnimStyle `BackAndForthContinuous`, driven by the `Wipers` control. The sample also plays a cab wiper animation (`CabView\sample_wipers.ia`) from the same control through an interior slave element.

verified

Doors (wagons and coaches)

One .ia per side (`doors_left` , `doors_right`) so the script can open the platform side only. Name leaves with the `door$$` keyword. Doors are usually toggled by a control (`DoorsOpenClose` exists on the steam sample) or by Lua with `AddTime` (Part 5).

7 · Project example: trolley pole with route-specific wire heights

The Texas Electric trolley-pole revision spec defines the pole position as **animation frames 0–100**, with wire-height presets at frames 65, 72, 78 and 92, and a gravity fall of 50 frames/s. Make the animation match the spec exactly so the Lua (Part 5) can use spec numbers directly:

1. Key the pole from roof saddle (frame 0) to maximum travel (frame 100). Frame numbers are then literally the spec's positions.
2. Set Blender to a round rate for this file's timing maths. At 100 fps, 1 frame = 0.01 s of animation time, so “move to frame F” = animation time F/100 s. (Check `SampleRate` in the log.)
3. Export two files (front and rear pole) with independent AnimSet IDs, e.g. `poleFront` and `poleRear`.
4. The trailing/leading swing and the 3rd hidden radial bogie in the spec are separate nodes with their own animation or bogie entry; keep them out of the pole's .ia.
5. For the stowed/locked mesh swap in the spec, model both meshes and let the blueprint's visibility elements (or `ActivateNode` from Lua) show one at a time. The static stowed mesh must be its own node.

> **Scrubbed exterior animation** check
>
> Exterior animations are played with `Call("*:AddTime", id, seconds)` and `Call("*:Reset", id)` (documented in the TS2015 reference; AddTime returns non-zero at the end). To hold the pole at an arbitrary frame, keep your own frame counter in Lua and add the difference each update. Many community scripts also use a `SetTime` call; it is not in the supplied reference, so test it in your build before relying on it.

8 · Exporting .ia files with BRIAGE, step by step

1. Export the IGS first, and **do not change object names or hierarchy** between IGS and IA exports.
2. In the IA panel (below the IGS panel in Scene properties) choose Export selected, **Relative export** on, **Selected hierarchy** on.
3. Select the animation's root objects. Include: objects without a parent that are animated; objects whose parent is an armature; parents of animated children (BRIAGE §8.5.5 / annex 10.5). Children of selected objects come along automatically.
4. Set the frame range to the animation (0–34 in the sample), or rely on Trim Animation.
5. Export to the matching folder: exterior animations next to the exterior IGS, cab animations in `CabView`.
6. Repeat per animation. With the toolkit, tag each root object with a custom property `tsc_ia = "sample_throttle"` and press **Prepare next IA job**: it selects the object, sets the range, rewinds to frame 0 and copies the file name to the clipboard.

9 · Troubleshooting

Symptom	Cause and fix
Nothing moves in the preview	AnimSet entry lacks “Animate in editor = True” (BRIAGE §9.5); or the node was merged (check the Groups list); or the .ia was exported from different names.
Part jumps to a wrong position before animating	Rest pose on frame 0 differs from the pose at IGS export; apply rotation/scale before keying; for stubborn children use a Child Of constraint with Set Inverse and re-export the IGS (BRIAGE §9.9).
Whole model slides during an animation	IA selection is inconsistent with the IGS structure (e.g. a child selected without its parent while Relative export is off) (BRIAGE §8.5.5).
Stuttering	Linear interpolation; raise Frame rate Multiplier; for needles use 7× and an odd frame count (BRIAGE §9.6, §8.5.4).
Preview still shows the old animation	Close the Asset Editor/preview, clear its cache, restart (BRIAGE §9.1.1).
Lever moves but is offset from the value	Non-linear keys, or keys not at the first/last frame, or control min/max not matching the keyed extremes.
Scale animation ignored	IA has no scale. Fake it with bones and skinning (BRIAGE §8.5.2) or swap meshes with visibility.